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Global Retail Lending in the Aftermath of the US Financial Crisis: Distinguishing between Supply and Demand Effects

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1 Big picture

- **Research question.** Did the US financial crisis reduce retail credit supply outside the United States? More specifically, did German savings banks that were indirectly exposed to US subprime losses through their Landesbank ownership reject more retail loan applications than comparable nonexposed savings banks?
- **Main idea.** The paper uses German savings banks as a laboratory for identifying the bank lending channel. Some local savings banks were hit by the crisis because they owned regional Landesbanken with subprime losses, even though the savings banks themselves operated locally and served local borrowers.
- **Key empirical advantage.** The authors observe the universe of loan applications and approvals at savings banks. This makes it possible to separate credit demand from credit supply: applications measure demand, while approval or rejection decisions measure supply.
- **Treatment.** Affected savings banks are those connected to Sachsen LB, Bayern LB, and West LB, the Landesbanken that were hit by the financial crisis during the sample period. Nonaffected savings banks are connected to other Landesbanken.
- **Main finding.** After August 2007, affected savings banks sharply reduce loan acceptance rates relative to nonaffected banks. The effect is large, statistically strong, and especially pronounced for mortgage loans.
- **Demand-side finding.** Loan applications fall after the crisis, but they fall similarly at affected and nonaffected banks. Loan amounts requested by mortgage applicants also do not fall significantly more at affected banks.
- **Relationship finding.** Existing bank-depositor relationships mitigate the supply contraction. Relationship borrowers are less likely to be rejected by affected banks after the crisis.
- **Economic takeaway.** A global financial shock can transmit to local retail credit markets through bank balance-sheet exposure, even when local economic conditions and loan demand do not differentially deteriorate.

2 Incremental contribution of the paper

- **From corporate lending to retail lending.** Much crisis work studies corporate credit. This paper studies household and retail credit, where borrowers have fewer alternative financing sources and direct application data are rarely available.
- **Direct separation of supply and demand.** The data include rejected applications, not only originated loans. This avoids the standard problem that observed lending combines borrower demand and bank supply.

- **Clean institutional setting.** German savings banks follow the regional principle: they lend only in their local municipality. This limits the scope for strategic nationwide expansion and makes local bank exposure easier to interpret.
- **Natural experiment through Landesbank ownership.** Exposure to the US subprime crisis is generated by savings banks' ownership of regional Landesbanken, not by the local borrowers themselves.
- **Borrower-level credit quality.** The data contain internal risk ratings for applicants, so the authors can control for observable borrower quality and check that the result is not driven by shifts in applicant risk.
- **Relationships in retail credit.** The paper shows that relationship lending matters not only for firms but also for retail customers during a crisis.
- **Global banking channel.** The paper provides evidence that financial globalization can affect domestic households through local banks' indirect balance-sheet exposure.

3 Institutional setting and shock

3.1 German savings banks and Landesbanken

German savings banks are local public banks. They are required to serve customers in their own municipality and generally do not lend outside that region. Their mandate is not simply profit maximization, but local financial access.

Landesbanken are regional public banks. They are jointly owned by federal states and savings bank associations. They serve three roles: house bank for the state, wholesale and clearing bank for savings banks, and commercial bank.

The important institutional link is vertical support. Savings banks own their Landesbank and are expected to support it when losses threaten solvency. Thus, when a Landesbank suffers losses on US subprime assets, its owner savings banks may face balance-sheet pressure even if their own retail loan portfolios are local.

3.2 Affected and nonaffected banks

The treated savings banks are those linked to Landesbanken that were affected by the US financial crisis during the sample period:

- **Sachsen LB** in Saxony;
- **Bayern LB** in Bavaria;
- **West LB** in North Rhine-Westphalia.

Savings banks linked to other Landesbanken are treated as nonaffected in the main specification. The paper later checks robustness to alternative event dates based on public announcements of subprime losses.

3.3 Why the setting is useful

The setting has three features that make the empirical design credible.

- **Local borrowers are not directly exposed to US subprime.** Germany had stable housing prices and relatively strong macroeconomic conditions during the sample period.
- **Savings banks are geographically constrained.** Affected and nonaffected banks lend to different local markets, but the regional principle prevents them from changing markets in response to the shock.
- **Loan applications are observed.** If households simply demanded less credit from affected banks, that would show up in application counts or requested loan amounts.

Interpretation: the shock is not that German households suddenly became worse borrowers in affected regions. The shock is that their local banks became more constrained through exposure to Landesbanken.

4 Data and measurement

4.1 Loan application data

The main dataset covers loan applications to German savings banks from July 2006 through June 2008.

- **Applications.** The sample contains 1,296,726 retail loan applications.
- **Loan types.** About 71.5 percent are consumer loans and 28.5 percent are mortgage loans.
- **Approval outcome.** The dependent variable in the supply tests is whether the bank accepts the application.
- **Loan amount.** Loan amount is available for mortgage applications; the mean requested amount is about 86,609 euros.
- **Risk rating.** Each applicant has an internal rating from the savings bank's scoring system. Higher numbers indicate worse borrower risk.
- **Relationship indicator.** The relationship variable equals one if the applicant had a checking account with the same bank before applying for the loan.
- **Demand proxy.** Weekly loan applications by bank are used as a demand proxy.

4.2 Timing

The main post-crisis indicator equals one after August 2007. This date captures the start of the crisis period for the Landesbank exposure in the main design.

The paper also uses alternative timing based on when losses became publicly observable: Sachsen LB in the third quarter of 2007, West LB in the fourth quarter of 2007, and Bayern LB in the first quarter of 2008.

4.3 Core variables

- **Affected.** Equals one for applications to savings banks that own affected Landesbanken.
- **Post-August 2007.** Equals one after August 2007.
- **Accepted.** Equals one if the loan application is approved.
- **Internal rating.** Bank-specific measure of borrower credit quality.
- **Relationships.** Equals one if the borrower had a checking account with the bank before applying.
- **Bank size and liquidity.** Used to examine whether weaker banks cut credit more.
- **Consumer confidence.** Controls for aggregate demand conditions.

5 Methodology and empirical specifications

5.1 Difference-in-differences for loan supply

The main supply test estimates whether affected banks changed approval behavior after August 2007 relative to nonaffected banks. A simplified version of the estimating equation is:

$$Y_{i,b,t} = A_b + B_t + \delta X_{i,b,t} + \beta_1(\text{Affected}_b \times \text{Post}_t) + \beta_2(\text{Nonaffected}_b \times \text{Post}_t) + \varepsilon_{i,b,t},$$

where $Y_{i,b,t}$ equals one if application i at bank b in period t is accepted.

The difference-in-differences effect is the difference between the post-crisis change for affected banks and the post-crisis change for nonaffected banks. Economically, the key question is whether β_1 is more negative than β_2 .

Controls include borrower internal rating, consumer confidence, bank fixed effects, and time fixed effects. The main estimates use linear probability models; probit specifications are used as robustness checks.

5.2 Demand tests

The paper tests whether lower observed lending reflects demand rather than supply using two outcomes.

- **Number of applications.** If customers in affected regions demand less credit, affected banks should receive fewer applications after the crisis.

- **Requested loan amount.** If mortgage borrowers at affected banks became more cautious, requested loan amounts should fall more at affected banks.

The results show a general decline in demand after the crisis, but not a differential decline at affected banks.

5.3 Relationship lending and triple differences

To test whether existing relationships protect borrowers during the crisis, the paper estimates a triple-difference specification. The key term is:

$$\text{Affected}_b \times \text{Post}_t \times \text{Relationships}_{i,b,t}.$$

A positive coefficient means that relationship borrowers at affected banks suffer less from the post-crisis supply contraction than new borrowers at those same banks.

5.4 Bank heterogeneity

The paper also asks which affected banks cut lending most. The key heterogeneity variables are:

- **Bank size.** Smaller banks are expected to be less able to absorb the shock.
- **Liquidity.** Banks with lower liquid assets are expected to cut lending more.
- **Loan type.** Mortgage loans may be more affected than consumer loans because they are larger, longer term, and more balance-sheet intensive.

6 Identification and interpretation

6.1 What the paper can identify

The paper identifies whether retail loan approval rates fell more at banks indirectly hit by the US financial crisis than at otherwise similar savings banks. Because applications are observed, the evidence speaks directly to supply-side credit rationing rather than merely to lower lending volumes.

6.2 Parallel-trend assumption

The difference-in-differences design requires that, absent the financial crisis, affected and nonaffected banks would have had similar trends in acceptance rates. The paper supports this assumption in three ways.

- Before August 2007, affected and nonaffected banks have similar acceptance rates.

- A placebo event in the pre-crisis period does not generate a differential effect.
- Geographic proximity tests compare affected banks with nearby nonaffected banks in neighboring states.

6.3 Demand versus supply

The central identification problem is separating bank credit supply from borrower demand. The paper's solution is to observe both applications and approval decisions.

- If demand changed, application counts or requested loan amounts should move differentially.
- If supply changed, rejection rates should move differentially conditional on applications and borrower risk.

The evidence points to supply: affected banks reject more applications, while demand proxies do not decline more at affected banks.

6.4 Local economic confounds

Affected Landesbanken are located in different German regions. A concern is that these regions may have different economic shocks. The paper addresses this by controlling for consumer confidence, comparing geographically close banks, and showing that borrower risk distributions do not change in a way that explains the findings.

6.5 Interpretation of relationships

Relationships can help for two reasons. First, banks may have soft information about existing customers, making these applicants less risky to approve during a crisis. Second, banks may value relationship continuation. The paper finds that relationship effects are stronger for consumer loans than mortgage loans, consistent with relationships mattering more when collateral is less central.

7 Tables and figures

7.1 Table 1 and Figure 1: Institutional exposure and aggregate lending

Read Table 1:

- Table 1 lists the 11 Landesbanken and their owners.
- The affected Landesbanken are Sachsen LB, Bayern LB, and West LB.

- Savings bank associations own large shares of these Landesbanken, so local savings banks are financially connected to Landesbank losses.

Read Figure 1:

- Panel A shows that aggregate bank lending in Germany continued to rise during the early crisis period.
- Panel B shows that savings-bank total and corporate lending continued to grow, while consumer lending declined.
- The figure motivates the core question: is the retail lending decline demand-driven or supply-driven?

Economic message: The institutional ownership link creates exposure, while aggregate lending patterns show that the crisis did not simply shut down all German lending. The paper therefore focuses on retail credit supply at affected local banks.

Table 1
Ownership structures of Landesbanken.
This table provides an overview of the German Landesbanken and their respective owners. Ownership by savings banks is denoted by (S). The upper part of the table shows the three Landesbanken that are affected by the financial crisis after August 2007 and during our sample period. The lower part shows the remaining eight Landesbanken. The information on the Landesbanken is provided by Bundesverband Öffentlicher Banken Deutschlands (VÖB) and represents the status as of May 2008.

Landesbank	Owner	Share (percent)
Affected		
Sachsen LB	Sachsen-Finanzgruppe (S)	62.960
(acquired by Landesbank Baden-Württemberg on March 6, 2008)		
Bayern LB	Freistaat Sachsen	37.040
	Freistaat Bayern	50.000
	Sparkassenverband Bayern (S)	50.000
West LB	NRW Bank	31.200
	Rheinischer Sparkassen- und Giroverband (S)	25.200
	Westfälisch-Lippischer Sparkassen- und Giroverband (S)	25.200
	Land Nordrhein-Westfalen	17.400
	Landschaftsverband Rheinland	0.500
	Landschaftsverband Westfalen-Lippe	0.500
Nonaffected		
Bremer Landesbank	Nord LB	92.500
Kreditanstalt Oldenburg-Girozentrale	Freie Hansestadt Bremen	7.500
ISH Nordbank	Freie und Hansestadt Hamburg	35.380
	Land Schleswig-Holstein	19.960
	Sparkassen- und Giroverband für Schleswig-Holstein (S)	18.050
Landesbank Baden-Württemberg	J.C. Flowers	26.610
	Land Baden-Württemberg	35.611
	Sparkassenverband Baden-Württemberg (S)	35.611
	Landeshauptstadt Stuttgart	18.932
	L-Bank, Landesbank Baden-Württemberg	4.923
	Sparkassen- und Giroverband Rheinland-Pfalz (S)	4.923
Landesbank Berlin	Landesbank Berlin Holding AG	100.000
Landesbank Hessen-Thüringen	Sparkassen- und Giroverband Hessen-Thüringen (S)	85.000
	Land Hessen	10.000
	Freistaat Thüringen	5.000
Landesbank Rheinland-Pfalz	Landesbank Baden-Württemberg	100.000
Norddeutsche Landesbank	Land Niedersachsen	41.750
	Land Sachsen-Anhalt	8.250
	Sparkassenverband Niedersachsen (S)	37.250
	Sparkassenbeteiligungsverband Sachsen-Anhalt	7.530
	Sparkassenbeteiligungsverband Mecklenburg-Vorpommern	5.220
Saar LB	Bayern LB	75.100
	Sparkassenverband Saar (S)	14.900
	Saarland	10.000

Table 1: Ownership structures of Landesbanken.

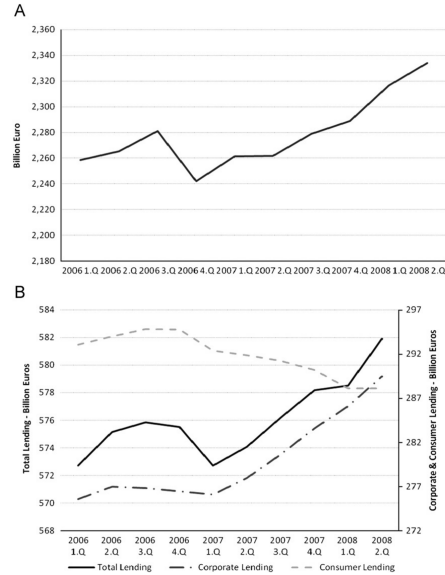


Figure 1: Aggregate lending in Germany. Panel A shows the aggregate lending (corporate and consumer) for all bank groups. Panel B shows the aggregate consumer and corporate lending for savings banks. Loan types do not include mortgage loans.

7.2 Figure 2 and Table 2: Geography and loan-application data

Read Figure 2:

- Figure 2 maps the geographic reach of the three affected Landesbanken.
- Affected regions include Saxony, Bavaria, and North Rhine-Westphalia.
- The map illustrates why geographic controls and robustness checks are important.

Read Table 2:

- Table 2 reports descriptive statistics for 1,296,726 loan applications.
- The average acceptance rate is 95.6 percent.
- About 36.5 percent of applications are to affected banks, and 49.3 percent are in the post-August 2007 period.
- Mortgage loans account for 28.5 percent of applications.
- The average internal rating is 6 on a 1–12 scale, and 94.1 percent of applicants have a prior relationship.

Economic message: The dataset is unusually powerful because it contains both accepted and rejected applications, risk ratings, loan type, relationship status, and bank-week demand measures.

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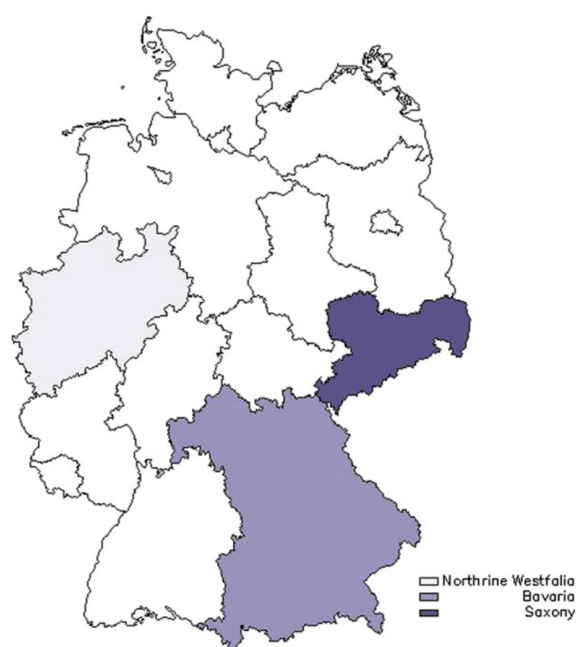


Fig. 2. Geographical reach of affected Landesbanken. The figure shows the geographical reach of the three Landesbanken that are affected by the financial crisis after August 2007 and during our sample period. They represent Westdeutsche Landesbank or West LB (North Rhine-Westphalia), Bayerische Landesbank or Bayern LB (Bavaria), and Landesbank Sachsen or Sachsen LB (Saxony).

Figure 2: Geographical reach of affected Landesbanken.

Table 2
Descriptive statistics.
This table presents summary statistics for the sample of loan applications from German savings banks from July 2006 through June 2008. The number of observations corresponds to the number of loan applications. Loan amounts are available only for the subset of mortgage loans. The number of weekly loan applications is presented in bank-week units.

	Number of observations	Mean	Standard deviation	p (25)	Median	p (75)	Minimum	Maximum
Inference variables								
Affected	1,296,726	0.365	0.481	0	0	1	0	1
Post-August 2007	1,296,726	0.493	0.500	0	0	1	0	1
Affected × Post-August 2007	1,296,726	0.180	0.385	0	0	0	0	1
Nonaffected × Post-August 2007	1,296,726	0.313	0.464	0	0	1	0	1
Loan type								
Mortgage loans	1,296,726	0.285	0.452	0	0	1	0	1
Dependent variables								
Accepted (Yes/No)	1,296,726	0.956	0.205	1	1	1	0	1
Loan amount	317,616	86,609	69,360	25,000	70,900	132,000	5,000	238,000
No. of weekly loan applications	33,885	40	76	11	18	40	11	927
Borrower characteristics								
Internal Rating	1,244,441	6	2.944	4	6	8	1	12
Relationship characteristics								
Relationships	1,296,726	0.941	0.235	1	1	1	0	1

Table 2: Descriptive statistics.

7.3 Tables 3 and 4: Main bivariate acceptance-rate evidence

Read Table 3:

- Before the crisis, aggregate acceptance rates are very similar for affected and nonaffected banks.

- After the crisis begins, affected banks' acceptance rates fall sharply.
- Nonaffected banks' acceptance rates remain close to their pre-crisis levels.

Read Table 4:

- Table 4 implements the basic difference-in-differences comparison.
- In the pooled sample, affected banks' acceptance rate falls from about 97.5 percent to 86.4 percent, while nonaffected banks remain around 97.6 percent.
- The pooled DID estimate is about 11.0 percentage points.
- The effect is large for consumer loans and even larger for mortgage loans.
- Panel D shows that the decline appears across internal rating classes, not only among the riskiest borrowers.

Economic message: The raw data already show the central supply-side result: affected banks reject substantially more applications after the crisis, while nonaffected banks do not.

Aggregate acceptance rates, affected versus nonaffected banks.

This table presents aggregate acceptance rates for affected versus nonaffected banks over time. Acceptance rates are aggregated across each quarter. The first Landesbank (Sachsen LB) was directly hit by the financial crisis in August 2007 (Q3-2007). At the same time, the massive exposure and vulnerability of the other Landesbanken (Bayern LB and West LB) also became obvious.

Quarter	Aggregate acceptance rate	
	Affected banks	Nonaffected banks
Q3-2006	97.34%	98.33%
Q4-2006	97.58	97.85
Q1-2007	97.75	97.67
Q2-2007	97.61	97.23
Q3-2007	93.96	97.52
Q4-2007	85.64	97.20
Q1-2008	84.58	97.53
Q2-2008	84.93	98.03

Table 3: Aggregate acceptance rates, affected versus nonaffected banks.

Table 4
Loan acceptance rates at the onset of the financial crisis (bivariate tests).
This table presents mean difference-in-differences (DID) estimates of loan acceptance rates. Savings banks are classified into two groups: affected, when savings banks are owner of one of the three Landesbanken that are affected by the US subprime mortgage crisis after August 2007; nonaffected, otherwise. Panels A through C report the results for the pooled sample (consumer and mortgage loans), only consumer loans, and only mortgage loans, respectively. Standard errors are reported in parentheses. The number of observations is reported in brackets. The DID estimate is printed in bold. Panel D reports mean DID estimates for the sample segregated by rating class. For brevity, we report only p-values of the difference and DID estimates. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	All	Affected	Nonaffected	Difference			
Panel A: Pooled consumer and mortgage loans							
Before August 2007	0.976 (0.0002) [657,309]	0.975 (0.0003) [239,644]	0.977 (0.0002) [417,665]	0.002*** (0.0004)			
After August 2007	0.943 (0.0003) [639,417]	0.864 (0.0007) [233,968]	0.976 (0.0002) [405,449]	0.113*** (0.0007)			
Difference	-0.041*** (0.0003)	-0.111*** (0.0008)	-0.001* (0.0003)	-0.110*** (0.000)			
Panel B: Consumer loans							
Before August 2007	0.978 (0.0002) [464,399]	0.979 (0.0003) [183,037]	0.977 (0.0003) [281,362]	0.002*** (0.0004)			
After August 2007	0.936 (0.0004) [462,426]	0.874 (0.0008) [186,971]	0.978 (0.0003) [275,455]	0.104*** (0.0008)			
Difference	-0.042*** (0.0004)	-0.105*** (0.0008)	0.001* (0.0004)	-0.105*** (0.000)			
Panel C: Mortgage loans							
Before August 2007	0.976 (0.0002) [192,910]	0.961 (0.0008) [56,607]	0.977 (0.0004) [136,303]	0.0163*** (0.0009)			
After August 2007	0.943 (0.0003) [176,991]	0.824 (0.0018) [46,997]	0.974 (0.0004) [129,994]	0.150*** (0.0018)			
Difference	-0.038*** (0.0007)	-0.137*** (0.0019)	-0.003*** (0.0006)	-0.134*** (0.000)			
Before August 2007		After August 2007					
Affected	Nonaffected	Difference (p-value)	Affected	Nonaffected	Difference (p-value)	Diff-in-Diff (p-value)	
Panel D: Diff-in-Diff by rating classes							
Borrower risk (Internal Rating)							
	0.986	0.993	0.007	0.876	0.993	0.117	0.110***
			-0.0001			<0.0001	<0.0001
2	0.988	0.989	(0.726)	0.889	0.989	0.100	0.099***
3	0.989	0.987	-0.002	0.898	0.989	0.091	0.093***
			(0.055)			<0.0001	<0.0001
4	0.990	0.988	-0.003	0.903	0.988	0.085	0.088***
			(0.001)			<0.0001	<0.0001
5	0.988	0.987	0.000	0.890	0.987	0.097	0.097***
			(0.607)			0.095	<0.0001
6	0.986	0.985	0.000	0.890	0.986	0.095	0.096***
			(0.429)			<0.0001	<0.0001
7	0.983	0.985	0.002	0.890	0.985	0.095	0.093***
			(0.046)			<0.0001	<0.0001
8	0.978	0.981	0.003	0.870	0.980	0.110	0.107***
			(0.005)			<0.0001	<0.0001
9	0.973	0.975	0.002	0.859	0.977	0.118	0.116***
			(0.285)			<0.0001	<0.0001
10	0.958	0.958	0.000	0.817	0.949	0.132	0.132***
			0.841			<0.0001	<0.0001
11	0.885	0.917	0.032	0.715	0.904	0.189	0.157***
			-0.0001			<0.0001	<0.0001
12	0.793	0.804	0.010	0.650	0.811	0.160	0.150***
			(0.107)			<0.0001	<0.0001

Table 4: Loan acceptance rates at the onset of the financial crisis.

7.4 Table 5: Multivariate loan-supply regressions

Read Table 5:

- Table 5 estimates linear probability models with bank and time fixed effects.
- The pooled DID effect is about 8.2 percentage points after controlling for borrower risk and other covariates.
- The consumer-loan DID effect is about 7.3 percentage points.
- The mortgage-loan DID effect is about 12.1 percentage points.
- The mortgage effect is significantly larger than the consumer-loan effect.

Economic message: The supply contraction survives borrower-risk controls and fixed effects. Affected banks do not merely receive worse applications; they approve fewer applications conditional on observable credit quality.

Table 5
Loan acceptance rates at the onset of the financial crisis (multivariate tests).
We estimate the probability that a bank accepts a loan application. All variables are defined in the Appendix. The borrower internal rating is the bank's internal risk assessment at the time the loan application is made. The models are estimated using a linear probability model with bank-specific fixed effects and year fixed effects. Panel A reports regression results for the pooled sample of consumer loans and mortgage loans. Panel B reports the results for the consumer loan and mortgage loan subsamples. Models 3, 4, and 5 further adjust the standard errors for possible autocorrelation at the bank level. The diagnostic section of Panel B reports the p-value from the Wald test under the H0 that, among the owners of the affected Ländersbanken, loan applicants for mortgage loans are as likely to be accepted as applicants for consumer loans after the start of the financial crisis. Interest, bank, and year fixed effects are not shown. Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Panel A: Pooled sample (consumer and mortgage loans)

	Consumer and mortgage loans					
	(1)		(2)		(3)	
(1) Affected + Post-August 2007	0.071***	(0.0006)	-0.072***	(0.0008)	-0.072***	(0.0027)
(2) Unaffected + Post-August 2007	0.011***	(0.0006)	0.019***	(0.0007)	0.019**	(0.0036)
Borrower risk (Internal Rating)						
1	0.228***	(0.0023)	0.228***	(0.0023)	0.228***	(0.0099)
2	0.218***	(0.0023)	0.218***	(0.0023)	0.218***	(0.0037)
3	0.209***	(0.0022)	0.209***	(0.0022)	0.209***	(0.0048)
4	0.207***	(0.0022)	0.207***	(0.0022)	0.207***	(0.0046)
5	0.203***	(0.0022)	0.203***	(0.0022)	0.203***	(0.0043)
6	0.200***	(0.0022)	0.200***	(0.0022)	0.200***	(0.0043)
7	0.197***	(0.0022)	0.197***	(0.0022)	0.197***	(0.0042)
8	0.194***	(0.0022)	0.194***	(0.0022)	0.194***	(0.0039)
9	0.182***	(0.0023)	0.182***	(0.0023)	0.182***	(0.0033)
10	0.177***	(0.0023)	0.177***	(0.0023)	0.177***	(0.0030)
11	0.097***	(0.0026)	0.097***	(0.0026)	0.097***	(0.0047)
Consumer Confidence						
Time fixed effects	Yes		Yes		Yes	
Bank fixed effects	Yes		Yes		Yes	
Standard errors clustered at bank level	Yes		Yes		Yes	
Diagnosics						
Adj. R ²	21.84%		21.84%		21.84%	
Wald Test: All coefficients = 0 (p-value)	< 0.0001		< 0.0001		< 0.0001	
Difference in differences						
DID estimate (1) - (2)	0.060***		0.060***		0.060***	
Wald test: (1) - (2) (p-value)	< 0.0001		< 0.0001		< 0.0001	
Number of observations	1,264,441		1,264,441		1,264,441	

Table 5, Panel A: Multivariate acceptance regressions, pooled sample.

Table 5, (continued)

Panel B: Consumer and mortgage loans

	Consumer loans			Mortgage loans		
	(4)	(5)	(6)	(7)	(8)	(9)
(1) Affected + Post-August 2007	-0.059***	(0.0009)	-0.068***	(0.0009)	-0.068***	(0.0010)
(2) Unaffected + Post-August 2007	0.014***	(0.0009)	0.017***	(0.0009)	0.017***	(0.0012)
Borrower risk (Internal Rating)						
1	0.222***	(0.0026)	0.222***	(0.0026)	0.215***	(0.0099)
2	0.215***	(0.0025)	0.215***	(0.0025)	0.205***	(0.0039)
3	0.212***	(0.0024)	0.212***	(0.0024)	0.198***	(0.0048)
4	0.210***	(0.0024)	0.210***	(0.0024)	0.196***	(0.0046)
5	0.207***	(0.0024)	0.207***	(0.0024)	0.193***	(0.0043)
6	0.205***	(0.0024)	0.205***	(0.0024)	0.191***	(0.0043)
7	0.203***	(0.0024)	0.203***	(0.0024)	0.189***	(0.0042)
8	0.198***	(0.0024)	0.198***	(0.0024)	0.185***	(0.0039)
9	0.189***	(0.0024)	0.189***	(0.0024)	0.177***	(0.0033)
10	0.182***	(0.0025)	0.182***	(0.0025)	0.169***	(0.0030)
11	0.096***	(0.0028)	0.096***	(0.0028)	0.086***	(0.0047)
Consumer Confidence						
Time fixed effects	Yes		Yes		Yes	
Bank fixed effects	Yes		Yes		Yes	
Standard errors clustered at bank level	Yes		Yes		Yes	
Diagnosics						
Adj. R ²	23.17%		23.18%		23.88%	
Wald test: All coefficients = 0 (p-value)	< 0.0001		< 0.0001		< 0.0001	
Difference in differences						
DID estimate (1) - (2)	0.073***		0.073***		0.121***	
Wald test: (1) - (2) (p-value)	< 0.0001		< 0.0001		< 0.0001	
Within treatment group						
(8) - (9)					0.055***	
Wald test: (8) - (9) (p-value)					< 0.0001	
Number of observations	926,825		926,825		317,616	

Table 5, Panel B: Consumer and mortgage loan subsamples.

7.5 Tables 6 and 7: Bank heterogeneity and demand through applications

Read Table 6:

- Table 6 asks whether the supply contraction is stronger for particular affected banks.
- Larger affected banks are more likely to continue accepting applications after the shock.
- More liquid affected banks are also more likely to continue lending.
- The results suggest that the balance-sheet channel is strongest for smaller and less liquid banks.

Read Table 7:

- Table 7 studies demand using the number of weekly applications per bank.
- Applications decline after August 2007 for both affected and nonaffected banks.
- The difference-in-differences estimates are insignificant.
- This weakens the view that the acceptance-rate decline is driven by lower borrower demand in affected regions.

Economic message: The supply contraction is concentrated where bank constraints should bind most. At the same time, demand falls generally but not differentially at affected banks.

Table 6
Loan acceptance rates and bank heterogeneity

We estimate the probability that a bank accepts a loan application. All variables are defined in the appendix. Log (Bank Size) is the natural logarithm of the bank's asset size in millions of euros. Liquidity (% of Total Assets) is the ratio of the bank's cash and marketable securities to total assets. The borrower internal rating is the bank's internal risk assessment at the time the loan application is made. The models are estimated using a linear probability model with year fixed effects. Intercept and year fixed effects are not shown. Heteroskedasticity consistent standard errors clustered at the bank level are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	Dependent variable: Accepted (Yes/No) Affected banks (post -August 2007)												
	Pooled sample (1)		Consumer loans (2)		Mortgage loans (3)		Pooled sample (4)		Consumer loans (5)		Mortgage loans (6)		
Log (Bank Size) (Liquidity (% of Total Assets))	0.049***	(0.016)	0.044***	(0.0156)	0.0794***	(0.0253)		16.149***	(6.311)	13.910***	(5.26)	31.866***	(11.2482)
<i>Borrower risk (Internal Rating)</i>													
1	0.227***	(0.046)	0.278***	(0.0415)	0.348***	(0.0694)	0.231***	(0.0481)	0.281***	(0.0429)	0.332***	(0.067)	
2	0.245***	(0.0416)	0.279***	(0.0409)	0.304***	(0.0666)	0.246***	(0.0427)	0.278***	(0.0418)	0.290***	(0.0633)	
3	0.254***	(0.0401)	0.267***	(0.0401)	0.288***	(0.067)	0.253***	(0.0411)	0.264***	(0.0409)	0.287***	(0.064)	
4	0.260***	(0.0403)	0.264***	(0.0401)	0.333***	(0.0676)	0.258***	(0.0411)	0.261***	(0.0409)	0.321***	(0.0645)	
5	0.250***	(0.0393)	0.253***	(0.0393)	0.307***	(0.0648)	0.247***	(0.0401)	0.249***	(0.0401)	0.294***	(0.0623)	
6	0.246***	(0.0388)	0.249***	(0.0386)	0.304***	(0.0658)	0.245***	(0.0394)	0.246***	(0.0394)	0.294***	(0.063)	
7	0.250***	(0.0391)	0.252***	(0.0392)	0.276***	(0.0591)	0.245***	(0.0398)	0.247***	(0.0399)	0.260***	(0.0569)	
8	0.229***	(0.0402)	0.230***	(0.0403)	0.268***	(0.0588)	0.225***	(0.0406)	0.225***	(0.0409)	0.256***	(0.0566)	
9	0.218***	(0.0381)	0.218***	(0.0379)	0.229***	(0.0677)	0.212***	(0.0385)	0.213***	(0.0384)	0.224***	(0.0641)	
10	0.173***	(0.0376)	0.174***	(0.0379)	0.156***	(0.0493)	0.168***	(0.0378)	0.170***	(0.0382)	0.146***	(0.0485)	
11	0.072***	(0.0263)	0.071***	(0.0266)	0.0716	(0.0445)	0.071***	(0.0268)	0.071***	(0.0272)	0.0570	(0.0439)	
Consumer Confidence	0.0135	(0.0099)	0.016*	(0.0096)	-0.0076	(0.0149)	0.0128	(0.0098)	0.0155	(0.0096)	-0.0082	(0.0149)	
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Standard errors clustered at bank level	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
<i>Diagnostics</i>													
Adj. R ²	5.13%		5.64%		6.95%		6.26%		6.44%		10.21%		
Wald test: All coefficients=0 (p-value)	< 0.0001		< 0.0001		< 0.0001		< 0.0001		< 0.0001		< 0.0001		
<i>Mortgage - consumer loans</i>													
Δ [Log Bank Size] or Δ [Liquidity (% of Total Assets)]					0.035***						17.965***		
p-value					< 0.0001						< 0.0001		
Number of observations	207,609		176,793		30,816		207,609		176,793		30,816		

Table 6: Loan acceptance rates and bank heterogeneity.

Table 7
Demand for loans after the onset of the financial crisis (applications).

The dependent variable is the number of loans requested by borrowers each week. We estimate the regressions for the pooled sample (consumer and mortgage loans, Columns 1 and 2) as well as consumer (Columns 3 and 4) and mortgage loans (Columns 5 and 6). The regressions are estimated using a fixed effect ordinary least squares (OLS) model and a negative binomial model with fixed effects to account for the nature of the dependent variable (count data). We further adjust the standard errors for possible autocorrelation at the state level. The diagnostic section of the table reports the difference-in-differences (DID) estimate as well as the p-value from the Wald test under the H0 that the DID estimate is equal to zero. The unit of our analysis is the number of weekly loan applications of each single bank and not an individual loan application. This reduces our sample size compared with Table 4 and Table 5. We therefore use the mean internal rating (averaging the internal rating score across all loan applications per bank in a given week) to control for borrower risk. When using the negative binomial model, we further report the likelihood ratio test and in each case reject the H0 that conditional mean and median of the number of weekly loan applications is identical indicating overdispersion. We further do not find an elevated number of zeros in the dependent variable and therefore do not report the regressions using either Poisson or the zero inflated Poisson model. Intercept, bank and year fixed effects are not shown. Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	Consumer and mortgage loans		Consumer loans		Mortgage loans	
	(1) OLS	(2) Negative Binomial	(3) OLS	(4) Negative binomial	(5) OLS	(6) Negative binomial
(1) Affected × Post-August 2007	-8.131** (3.5957)	-0.207** (0.0896)	-8.133** (3.254)	-0.189** (0.0895)	-10.4366 (5.9764)	-0.244** (0.1161)
(2) Unaffected × Post-August 2007	-9.749*** (2.918)	-0.284*** (0.0514)	-10.753*** (2.1651)	-0.291*** (0.0429)	-12.2409 (5.83)	-0.257*** (0.072)
Mean internal rating	-1.245** (0.635)	-0.039*** (0.015)	-1.423*** (0.4195)	-0.0682*** (0.0159)	-0.3932 (0.8432)	0.0128 (0.0189)
Consumer Confidence	0.878* (0.4289)	0.023*** (0.0045)	0.3830 (0.3657)	0.020*** (0.0038)	1.482** (0.5786)	0.0245** (0.0068)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at state level	Yes	Yes	Yes	Yes	Yes	Yes
<i>Diagnostics</i>						
Adj. R ² / Pseudo-R ²	80.98%	22.10%	81.41%	22.03%	86.05%	23.04%
LR-test: $\alpha = 0$ (p-value)		< 0.0001		< 0.0001		< 0.0001
Difference-in-differences	1.6180	0.0770	2.6300	0.1010	1.8124	0.0130
Wald test: (1) - (2) [p-value]	0.6599	0.4293	0.4581	0.3319	0.7939	0.7907
Number of observations	32,638	32,638	25,822	25,822	6,816	6,816

Table 7: Demand for loans after the onset of the crisis, applications.

7.6 Tables 8 and 9: Loan amounts and relationship lending

Read Table 8:

- Table 8 uses requested mortgage loan amounts as another demand proxy.
- Loan amounts decline after August 2007, but not significantly more at affected banks.
- This again suggests that borrower demand does not explain the supply result.

Read Table 9:

- Table 9 tests whether bank-borrower relationships mitigate crisis-time rationing.
- Existing relationships are positively associated with approval probabilities.
- The triple interaction Affected × Post × Relationships is positive and significant.
- Relationship borrowers at affected banks are about 4.9 percentage points more likely to be approved after the crisis relative to comparable new borrowers.
- The relationship effect is stronger for consumer loans than for mortgage loans.

Economic message: Relationships provide insurance against bank credit rationing. Soft information and ongoing bank-customer ties matter when banks become constrained.

Table 8

The demand for loans after the onset of the financial crisis (loan amount).
The dependent variable is the natural logarithm of the loan amount requested by the borrower. Loan amounts are available only for the subset of mortgage loans in our sample that corresponds to 317,583 loans. The borrower internal rating is the bank's internal risk assessment at the time the loan application is made. The regressions are estimated using a conditional fixed effect ordinary least squares (OLS) model. Column 3 further adjusts the standard errors for possible autocorrelation at the state level. The diagnostic section of the table reports the DID estimate as well as the p-value from the Wald test under the H0 that the difference-in-differences (DID) estimate is equal to zero. The unit of our analysis is a loan application and not only an accepted loan. Intercept, bank, and year fixed effects are not shown. Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	(1) OLS	(2) OLS	(3) OLS
(1) Affected > Post-August 2007	-0.049*** (0.0091)	-0.049*** (0.0092)	-0.0490 (0.0203)
(2) Nonaffected > Post-August 2007	-0.045*** (0.0063)	-0.044*** (0.0066)	-0.0444 (0.0211)
Borrower risk (Internal Rating)			
1	-0.595*** (0.0146)	-0.595*** (0.0146)	-0.595*** (0.0268)
2	-0.449*** (0.0154)	-0.449*** (0.0154)	-0.449*** (0.0133)
3	-0.455*** (0.0154)	-0.455*** (0.0154)	-0.455*** (0.0101)
4	-0.346*** (0.0152)	-0.346*** (0.0152)	-0.346*** (0.0359)
5	-0.298*** (0.0151)	-0.298*** (0.0151)	-0.298*** (0.0324)
6	-0.192*** (0.0151)	-0.192*** (0.0151)	-0.192*** (0.025)
7	-0.117*** (0.015)	-0.117*** (0.015)	-0.117*** (0.022)
8	-0.087*** (0.015)	-0.087*** (0.015)	-0.087*** (0.0117)
9	-0.048*** (0.0149)	-0.048*** (0.0149)	-0.048*** (0.0112)
10	-0.041*** (0.015)	-0.041*** (0.015)	-0.041*** (0.0134)
11	-0.087*** (0.0159)	-0.087*** (0.0159)	-0.087*** (0.0106)
Consumer Confidence		-0.0004 (0.0012)	-0.0004 (0.0017)
Time fixed effects	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes
Standard errors clustered at state level			Yes
Diagnostics			
Adj. R ²	14.17%	14.17%	14.17%
Difference-in-differences			
DID estimate: (1) - (2)	0.0046	0.0046	0.0046
Wald test: (1) - (2) [p-value]	0.5762	0.5729	0.5953
Number of observations	317,583	317,583	317,583

Table 8: Demand for loans after the onset of the crisis, loan amount.

Table 9

Bank-borrower relationships during the financial crisis.

We estimate the probability that a bank accepts a loan application. All variables are defined in the Appendix. The borrower internal rating is the bank's internal risk assessment at the time the loan application is made. The models are estimated using a linear probability model with bank specific fixed effects and year fixed effects. Intercept, bank, and year fixed effects are not shown. Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	Dependent variable: Accepted (Yes/No)					
	Pooled sample (1) LPM		Consumer loans (2) LPM		Mortgage loans (3) LPM	
Secular effects						
Post-August 2007	0.011*** (0.0006)		0.014*** (0.0006)		0.006*** (0.0012)	
Relationships	0.028*** (0.0019)		0.009*** (0.0019)		0.018*** (0.0025)	
Second level interactions						
Affected > Post-August 2007	-0.081*** (0.0008)		-0.072*** (0.0008)		-0.119*** (0.0022)	
Relationships > Post-August 2007	0.004*** (0.0022)		0.007* (0.004)		-0.007*** (0.0027)	
Relationships > Affected	0.020*** (0.003)		0.016*** (0.004)		0.048*** (0.005)	
Diff-in-Diff-in-Diff						
Affected > Post-August 2007 > Relationships	0.049*** (0.005)		0.041*** (0.007)		0.018*** (0.008)	
Borrower risk (Internal Rating)						
1	0.221*** (0.0023)		0.218*** (0.0026)		0.206*** (0.0059)	
2	0.210*** (0.0023)		0.212*** (0.0025)		0.199*** (0.0059)	
3	0.204*** (0.0022)		0.209*** (0.0024)		0.184*** (0.0059)	
4	0.202*** (0.0022)		0.207*** (0.0024)		0.182*** (0.0059)	
5	0.198*** (0.0022)		0.203*** (0.0024)		0.172*** (0.0059)	
6	0.195*** (0.0022)		0.202*** (0.0024)		0.162*** (0.0059)	
7	0.193*** (0.0022)		0.200*** (0.0024)		0.151*** (0.0059)	
8	0.187*** (0.0022)		0.196*** (0.0024)		0.131*** (0.0059)	
9	0.180*** (0.0023)		0.188*** (0.0024)		0.128*** (0.0059)	
10	0.155*** (0.0023)		0.162*** (0.0025)		0.109*** (0.006)	
11	0.097*** (0.0026)		0.096*** (0.0028)		0.088*** (0.0064)	
Time fixed effects	Yes		Yes		Yes	
Bank fixed effects	Yes		Yes		Yes	
Diagnostics						
Adj. R ²	22.04%		23.25%		24.07%	
Wald test: All coefficients = 0 (p-value)	< 0.0001		< 0.0001		< 0.0001	
Mortgage - consumer loans						
Δ(Affected > Post-August 2007)					0.047***	
p-value					< 0.0001	
Δ(Affected > Post-August 2007 > Relationships)					-0.023***	
p-value					< 0.0001	
Number of observations	1,244,441		926,825		317,616	

Table 9: Bank-borrower relationships during the financial crisis.

7.7 Tables 10 and 11: Model robustness and geographic proximity

Read Table 10:

- Table 10 compares the linear probability model with probit specifications.
- The central finding remains: affected banks reduce credit supply after the crisis.
- The result is not an artifact of the linear probability model.

Read Table 11:

- Table 11 compares affected banks with geographically proximate nonaffected banks.
- This helps address the concern that treated regions had different local economic shocks.
- The estimates remain similar or stronger, supporting the interpretation that Landesbank exposure drives the result.

Economic message: The main finding is robust to alternative functional forms and to a tighter local comparison group.

Table 10

Compare models for loan supply.

We estimate the probability that a bank accepts a loan application. All variables are defined in the Appendix. This table shows that results from the linear probability model are robust to probit as alternative assumption about the distribution of the error term. We compare the linear probability model and the probit model (with and without fixed effects). We only show the coefficients for the linear probability model as well as the marginal effects for the probit model, respectively, for the interaction of affected and nonaffected banks with post-August 2007. Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	Dependent variable: Accepted (Yes/No)		
	Pooled sample	Consumer loans	Mortgage loans
Linear probability model			
(1) Affected × Post-August 2007	-0.071*** (0.0008)	-0.059*** (0.0009)	-0.116*** (0.0022)
(2) Nonaffected × Post-August 2007	0.010*** (0.0006)	0.014*** (0.0008)	0.005*** (0.0012)
Diff-in-Diff (p-value)	< 0.0001***	< 0.0001***	< 0.0001***
Probit (without fixed effects)			
(1) Affected × Post-August 2007	-0.081*** (0.0015)	-0.074*** (0.0016)	-0.095*** (0.0035)
(2) Nonaffected × Post-August 2007	0.006*** (0.0006)	0.011*** (0.0006)	0.004*** (0.0013)
Diff-in-Diff (p-value)	< 0.0001***	< 0.0001***	< 0.0001***
Probit (with fixed effects)			
(1) Affected × Post-August 2007	-0.034*** (0.0009)	-0.025*** (0.0009)	-0.067*** (0.0031)
(2) Nonaffected × Post-August 2007	0.004*** (0.0003)	0.005*** (0.0003)	0.001*** (0.0007)
Diff-in-Diff (p-value)	< 0.0001***	< 0.0001***	< 0.0001***

Table 10: Comparing linear probability and probit models.

Table 11

Geographic proximity and access to credit.

We estimate the probability that a bank accepts a loan application. All variables are defined in the Appendix. The borrower internal rating is the bank's internal risk assessment at the time the loan application is made. The models are estimated using a linear probability model (LPM) with bank pair-specific fixed effects in Regressions 1 to 3 and a probit model with standard errors clustered at the bank pair level in Regression 4. Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Dependent variable	Pooled sample, LPM, Accepted (Yes/No) (1)	Consumer loans, LPM, Accepted (Yes/No) (2)	Mortgage loans, LPM, Accepted (Yes/No) (3)	Pooled sample, Probit, Accepted (Yes/No) (4)
(1) Affected × Post-August 2007	-0.144*** (0.0022)	-0.114*** (0.0024)	-0.267*** (0.0055)	-0.149*** (0.0737)
(2) Nonaffected × Post-August 2007	0.007*** (0.0024)	0.011*** (0.0025)	0.015*** (0.006)	0.0052 (0.0146)
Affected	-0.009*** (0.0016)	-0.024*** (0.0017)	0.021*** (0.0049)	-0.0174 (0.0156)
Borrower risk (Internal Rating)				
1	0.145*** (0.0034)	0.185*** (0.0053)	0.187*** (0.0138)	0.041*** (0.0091)
2	0.165*** (0.0039)	0.187*** (0.0045)	0.186*** (0.0142)	0.042*** (0.0074)
3	0.168*** (0.0035)	0.186*** (0.0046)	0.172*** (0.0143)	0.047*** (0.0082)
4	0.172*** (0.0033)	0.177*** (0.0034)	0.186*** (0.0144)	0.051*** (0.0087)
5	0.168*** (0.0033)	0.172*** (0.0033)	0.178*** (0.0144)	0.051*** (0.0087)
6	0.169*** (0.0033)	0.173*** (0.0033)	0.163*** (0.0146)	0.053*** (0.0089)
7	0.168*** (0.0033)	0.171*** (0.0033)	0.160*** (0.0149)	0.053*** (0.0092)
8	0.154*** (0.0033)	0.156*** (0.0033)	0.137*** (0.0153)	0.048*** (0.0079)
9	0.147*** (0.0034)	0.148*** (0.0034)	0.127*** (0.0159)	0.045*** (0.0076)
10	0.125*** (0.0035)	0.128*** (0.0035)	0.059*** (0.017)	0.038*** (0.0077)
11	0.066*** (0.0038)	0.067*** (0.0037)	0.046*** (0.0188)	0.022*** (0.0054)
Consumer Confidence	-0.0010 (0.0004)	-0.0005 (0.0004)	-0.0002 (0.0001)	-0.0011 (0.0058)
Time fixed effects	Yes	Yes	Yes	Yes
Pairwise fixed effects	Yes	Yes	Yes	Yes
Standard errors clustered at bank pair	Yes	Yes	Yes	Yes
Diagnostics				
Adj. R ²	10.30%	9.87%	17.80%	19.24%
Number of Clusters	31	31	30	31
Wald test: All coefficients = 0 (p-value)	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Difference-in-differences				
DiD estimate: (1) - (2)	0.151*** (0.0001)	0.103*** (0.0001)	0.252*** (0.0001)	0.154*** (0.0001)
Wald test: (1) - (2) (p-value)	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Number of observations	206,083	171,901	34,182	206,083

Table 11: Geographic proximity and access to credit.

7.8 Table 12 and Appendix Table A1: Alternative event timing and variable definitions

Read Table 12:

- Table 12 uses public announcements of Landesbank subprime losses as event dates.
- The results are very similar to the main August 2007 specification.
- This shows that the supply result is not sensitive to one particular event-date definition.

Read Table A1:

- Table A1 defines the main inference variables, dependent variables, borrower characteristics, and relationship variables.
- It clarifies that the key outcome is application acceptance and that relationship status is based on a prior checking account at the same bank.

Economic message: The robustness evidence supports the interpretation that the crisis affected retail lending through bank exposure rather than through arbitrary timing or variable definitions.

Table 12

Public announcements of subprime losses.

We estimate the probability that a bank accepts a loan application. All variables are defined in the Appendix. The borrower internal rating is the bank's internal risk assessment at the time the loan application is made. The models are estimated using a linear probability model with bank-specific fixed effects and year fixed effects. In Models 1 to 4 we include Sachsen LB, West LB, and Bayern LB in the group of affected banks and identify the following quarters in which the announcements have been made: Sachsen LB: Q3-2007, West LB: Q4-2007, and Bayern LB: Q1-2008. In Models 5 to 8, we also include HSH Nordbank and LBFW in the group of affected Landesbanken and define Q2-2008 as the event date when they reported losses. We report only the difference-in-differences (DID) estimate (Affected > Post Public) and triple-diff estimate (Relationships > Affected > Post Public). Heteroskedasticity consistent standard errors are shown in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

	Pooled (1)	Consumer (2)	Mortgages (3)	Pooled (4)	Pooled (5)	Consumer (6)	Mortgages (7)	Pooled (8)
Affected > Post Public	-0.064*** (0.0177)	-0.055*** (0.0158)	-0.105*** (0.0346)	-0.102*** (0.0045)	-0.050*** (0.0147)	-0.043*** (0.0133)	-0.079*** (0.0248)	-0.072*** (0.0038)
Relationships > Affected > Post Public				0.039*** (0.0045)				0.021*** (0.0038)
Internal rating	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Consumer Confidence	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Diagnostics								
Adj. R ²	21.45%	22.74%	23.36%	21.60%	21.27%	22.59%	23.04%	21.41%
Wald test: all coefficients = 0 (p-value)	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Number of observations	1,244,441	926,825	317,616	1,244,441	1,244,441	926,825	317,616	1,244,441

Table 12: Public announcements of subprime losses.

Table A1
Main variables definitions.

Variable	Definition
Inference variables	
Affected	Dummy variable equal to one if the application is made to a savings bank that owns one of the affected Landesbanken (Bayern LB, Sachsen LB, West LB).
Post-August 2007	Dummy variable equal to one if the loan application is made after August 2007.
Affected > Post-August 2007	Interaction term: dummy variable equal to one if loan application is made to an affected savings bank after August 2007.
Nonaffected > Post-August 2007	Interaction term: dummy variable equal to one if loan application is made to a nonaffected savings bank after August 2007.
Loan type	
Mortgage Loans	Dummy variable equal to one if the loan type is a mortgage loan.
Dependent variable	
Accepted (Yes/No)	Dummy variable equal to one if the loan application was accepted.
Loan amount	The loan amount requested by the borrower in euros. The loan amount is available only for the subset of mortgage loans.
No. of weekly loan applications	Number of loan applications per bank per week.
Control variable	
Consumer Confidence	Measured by the Konsumklimaindex (consumer climate index) of the market research company C&K and is based on a monthly survey of two thousand consumers of age 14 and above. The index contains questions about how much consumers expect the economy and their income to grow and what they plan to consume. One sample question is: "Do you think it is currently advisable to spend a lot of money for consumption?" Three answers are possible: advisable, neutral, and not advisable. The answers are transferred into numbers and aggregated for all parts. The index long-term average is about zero.
Borrower characteristics	
Internal Rating	Based on a quantitative score that uses a scorecard at the loan application stage to facilitate and standardize the credit decision process across all savings banks. This credit score adds up individual scores based on age, occupation (for example, nature of an applicant's job and years the applicant has been in the job), and monthly repayment capacity based on the borrower's available income. The score also contains information on the existence and use of the borrower's checking and other accounts, i.e., credit or debit cards, the use of credit lines, and assets held in the bank. Based on past defaults of borrowers with similar characteristics, this score is consolidated into an internal credit rating, which is associated with a default probability for the borrower.
Relationships characteristics	
Relationships	Dummy variable equal to one if the loan applicant had a checking account with the same bank before the application. The regional principle excludes the possibility that a borrower has relationships with multiple sample banks.

Table A1: Main variable definitions.

8 Short summary

- The paper studies whether the US financial crisis reduced retail loan supply in Germany.
- The setting is German savings banks, which are local lenders but owners of regional Landesbanken.
- Some Landesbanken had large US subprime exposure, creating a shock to their owner savings banks.
- The authors observe the universe of loan applications and approvals from July 2006 through June 2008.
- Affected and nonaffected banks had similar acceptance rates before the crisis.
- After August 2007, affected banks reject substantially more applications.
- The effect remains after controlling for internal borrower risk, consumer confidence, bank fixed effects, and time fixed effects.
- The supply contraction is stronger for smaller banks, less liquid banks, and mortgage loans.
- Demand falls after the crisis, but it does not fall significantly more at affected banks.
- Existing bank-depositor relationships reduce the probability of rejection at affected banks after the crisis.
- Robustness checks using probit models, geographic proximity, alternative event timing, placebo trends, and borrower-quality tests support the main interpretation.

9 Conclusion

9.1 Core conclusion

Puri, Rocholl, and Steffen show that the US financial crisis transmitted to local retail credit markets in Germany through the banking system. Savings banks exposed through their Landesbank ownership cut loan supply to households relative to nonaffected savings banks.

9.2 Economic intuition

Affected banks faced a balance-sheet shock. Even though local borrowers were not directly responsible for the shock and even though loan demand did not differentially collapse, affected banks became more selective. This is the bank lending channel operating through local retail credit.

9.3 Incremental contribution revisited

The paper’s key contribution is the direct separation of supply and demand. Because rejected applications are observed, the authors can show that lower lending is not simply lower borrowing demand. The paper therefore provides unusually clean evidence of credit rationing in retail lending.

9.4 Implications

The findings imply that financial globalization can affect households through indirect banking exposures. Local banks can transmit international shocks even when they operate locally. The results also imply that bank-borrower relationships have real value during crises.

9.5 Limitations

The design studies German savings banks during a specific crisis episode. The estimates apply most directly to settings with relationship-based local banks and institutional exposure through bank ownership structures. Treatment is not randomly assigned, so the interpretation relies on the validity of the comparison between affected and nonaffected savings banks.

9.6 Takeaway

The enduring takeaway is that bank balance-sheet shocks matter for retail borrowers. When banks become constrained, they ration credit, especially in large and balance-sheet-intensive loans, but prior relationships can partially shield borrowers from the contraction.